trajectory of elecron

```
<< Graphics`Graphics3D`
```

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```
fz =
  Compile[{{x, _Real, 1}, {dt, _Real}},
   Module[{},
    Return[x - x / Sqrt[x . x] * dt +
      Sqrt[dt] *
       Table[Sum[Random[Real, {0, 1}], {12}] -
        6, {Length[x]}]]]]
```

```
CompiledFunction[{x, dt},
   Module[{},
    Return[x - (x dt)/Sqrt[x . x] + Sqrt[dt] Table[
       Sum[Random[Real, {0, 1}], {12}] - 6, {Length[x]}]],
   -CompiledCode-]
```

```
dt := 0.05
```

```
n := 5002
```

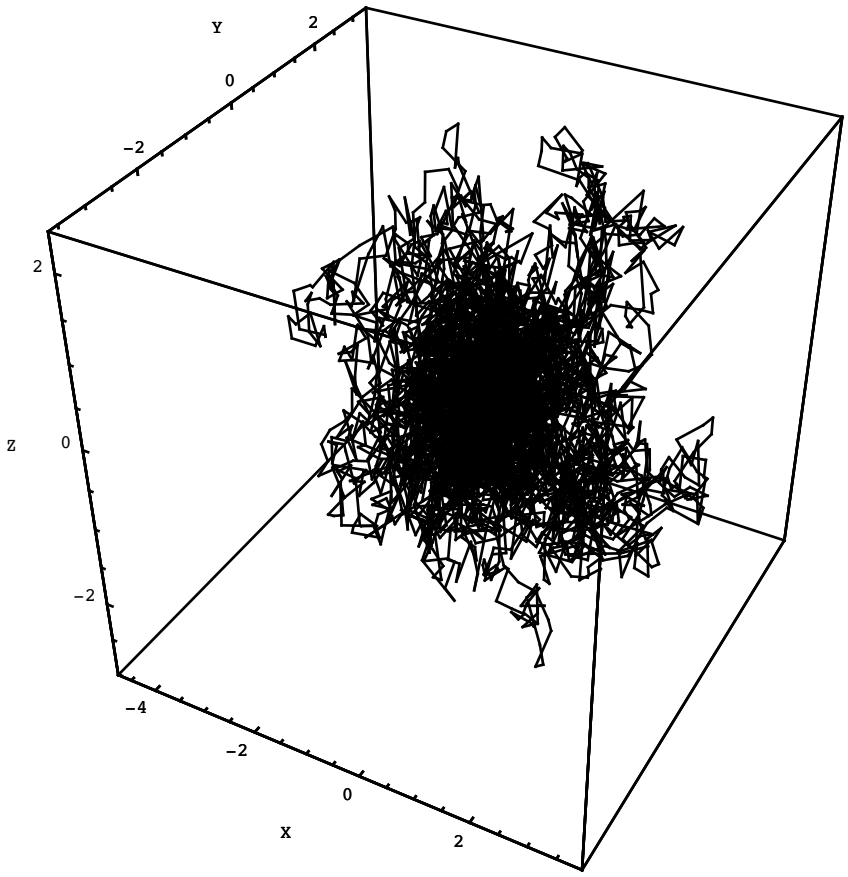
```
x = Table[{0.0, 0.0, 0.0}, {n}];
```

```
x[[1]] = {1.0, 1.0, 1.0};
```

```
Timing[
   Do[x[[i]] = fz[x[[i - 1]], dt], {i, 2,
     n}]][[1]]
```

```
0.203241 Second
```

```
ScatterPlot3D[x, PlotJoined → True,
    AxesLabel → {"\<X\>", "\<Y\>", "\<Z\>"},
    BoxRatios → {1, 1, 1},
    ColorFunction → {Hue[0.4]}];
```



```
Options[ScatterPlot3D]
```

```
{PlotJoined → False, PlotStyle → GrayLevel[0],
   AmbientLight → GrayLevel[0], AspectRatio → Automatic,
   Axes → True, AxesEdge → Automatic, AxesLabel → None,
   AxesStyle → Automatic, Background → Automatic,
   Boxed → True, BoxRatios → Automatic,
   BoxStyle → Automatic, ColorOutput → Automatic,
   DefaultColor → Automatic,
   DefaultFont :→ $DefaultFont,
   DisplayFunction :→ $DisplayFunction, Epilog → {},
   FaceGrids → None, FormatType :→ $FormatType,
   ImageSize → Automatic, Lighting → True,
   LightSources → {{{1., 0., 1.},
      RGBColor[1, 0, 0]}, {{1., 1., 1.},
      RGBColor[0, 1, 0]}, {{0., 1., 1.}, RGBColor[0, 0, 1]}},
   Plot3Matrix → Automatic, PlotLabel → None,
   PlotRange → Automatic, PlotRegion → Automatic,
   PolygonIntersections → True, Prolog → {},
   RenderAll → True, Shading → True,
   SphericalRegion → False, TextStyle :→ $TextStyle,
   Ticks → Automatic, ViewCenter → Automatic,
   ViewPoint → {1.3, -2.4, 2.},
   ViewVertical → {0., 0., 1.}}
```

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Modified for *Mathematica* 5 on 9 September 2003

```
fz =
  Compile[{{y, _Real, 2}, {dt, _Real}},
   Module[{x},
   x = y;
   Do[
     x⟦i + 1⟧ =
      x⟦i⟧ + ({1 / x⟦i, 1⟧, 0.0, 0.0} -
          0.5 * x⟦i⟧ / Sqrt[x⟦i⟧ . x⟦i⟧]) * dt +
       Sqrt[dt] *
        Table[Sum[Random[Real, {0, 1}], {12}] -
          6, {3}]
    , {i, Length[x] - 1}];
    Return[x]
  ]]
CompiledFunction[{y, dt},
   Block[{$$77},
    Module[{x}, x = y;
     Do[x⟦i + 1⟧ = ($$77 =
         x⟦
          i⟧; $$77 + ({1/x⟦i, 1⟧, 0.,
            0.} - 0.5 $$77/Sqrt[$$77 . $$77]) dt +
        Sqrt[dt] Table[Sum[Random[Real, {0, 1}], {12}] - 6, {3}]), {i,
       Length[x] - 1}]; Return[x]]], -CompiledCode-]

dt := 0.01

n := 100 000

dn := n / 2500

x = Table[{0.0, 0.0, 0.0}, {n}];

y = Table[{0.0, 0.0, 0.0}, {n}];

x⟦1⟧ = {-1. , 0. , 0. };

y⟦1⟧ = {1. , 0. , 0. };

Timing[x = fz[x, dt]]⟦1⟧
2.60502 Second
```

```
Timing[y = fz[y, dt]][[1]]
```

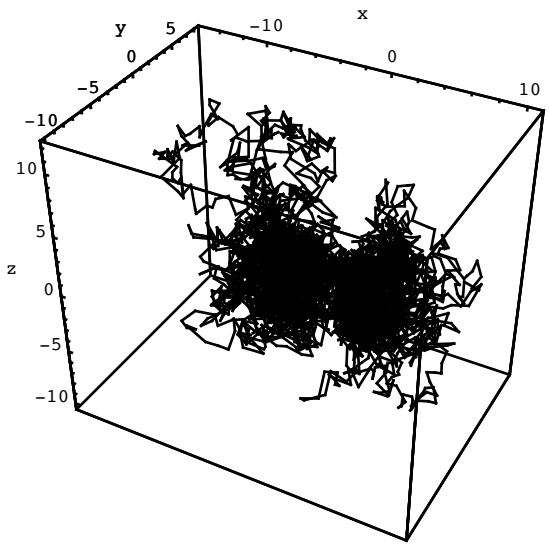
2.60121 Second

```
x1 = Table[x[[i]], {i, 1, Length[x], dn}];
```

```
y1 = Table[y[[i]], {i, 1, Length[y], dn}];
```

```
z = Join[x1, y1];
```

```
ScatterPlot3D[z, PlotJoined → True,
    AxesLabel → {"\<x\>", "\<y\>", "\<z\>"},
    ViewPoint → {1.300, -2.400, 2.000}];
```



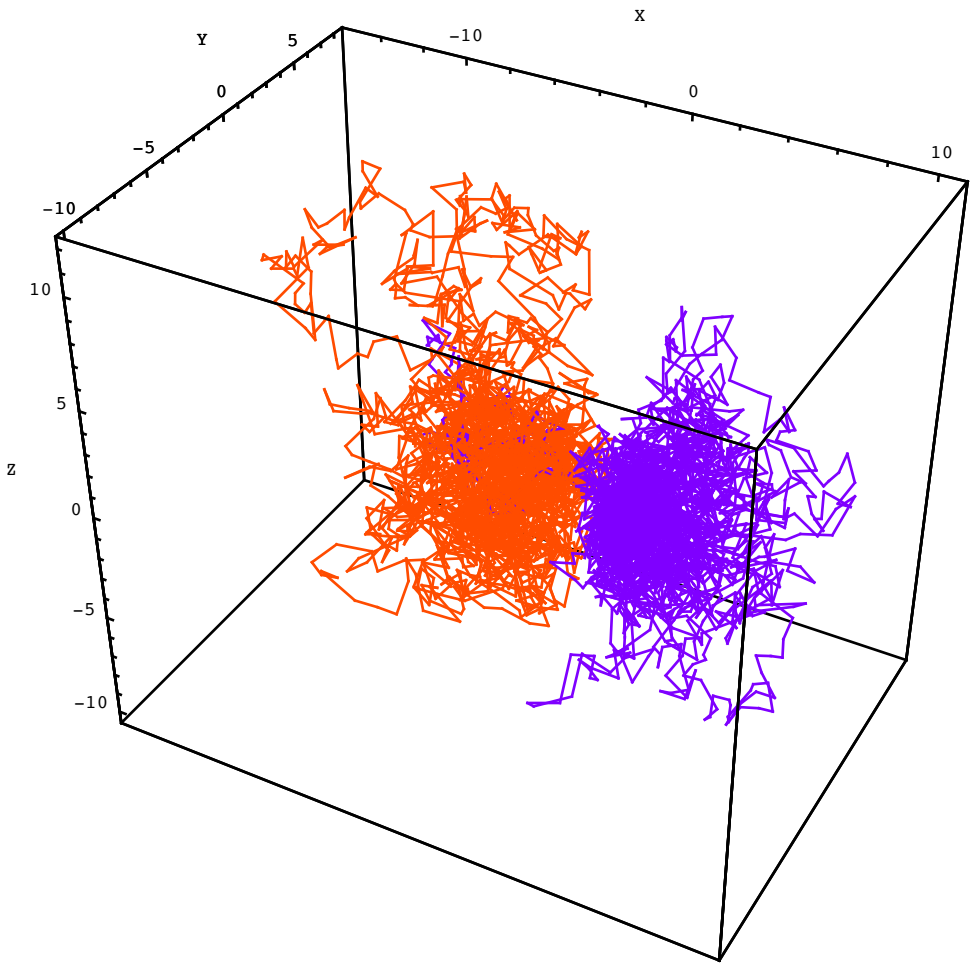

```
g1 = Graphics3D[{Hue[0.05], Line[x1]}]
```

- Graphics3D -

```
g2 = Graphics3D[{Hue[0.75], Line[y1]}]
```

- Graphics3D -

```
Show[{g1, g2}, Axes → True,
    AxesLabel → {"\<X\>", "\<Y\>", "\<Z\>"}];
```



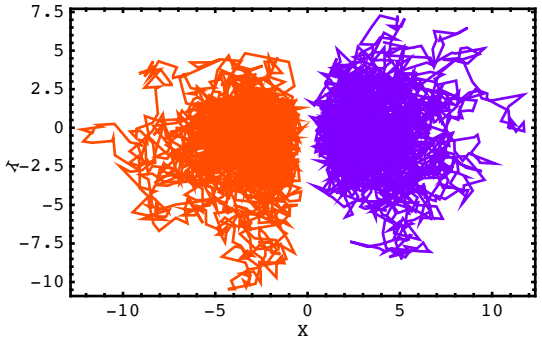

```
g3 =
   Graphics[{Hue[0.05],
     Line[Table[{x1[[i, 1]], x1[[i, 2]]}, {i,
        Length[x1]}]]}]
```

- Graphics -

```
g4 =
   Graphics[{Hue[0.75],
     Line[Table[{y1[[i, 1]], y1[[i, 2]]}, {i,
        Length[y1]}]]}]
```

- Graphics -

```
Show[g3, g4, Frame → True, Axes → False,
    FrameLabel → {"\<X\>", "\<Y\>"}];
```



References

Excel de Manabu Ryoushi Rikigaku (Learning about Quantum Mechanics with Excel),
Kunio Yasue, Blue Backs, Kodansha, 2001
(in japanese)

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